

PROJECT SUBMISSION • MILESTONE 2

ShopSense: Multi-Vendor Analytics Platform

Decision Intelligence, Inventory Forecasting & Multi-Tier Analytics

Welcome to the Milestone 2 presentation for ShopSense, an enterprise multi-vendor e-commerce decision and analytics platform. Building upon the relational database architecture, dual-token HTTP-Only cookie authentication, and Role-Based Access Control established in Milestone 1, Milestone 2 advances the platform into decision intelligence. This milestone introduces real-time warehouse inventory tracking with automated safety stock warnings, a 30-day predictive demand run-rate forecasting model, SQL-based RFM behavioral customer segmentation, and a multi-tiered rule-based product recommendation engine. The entire platform is built on a decoupled architecture separating FastAPI HTTP routers from dedicated Python service classes (app/services/), paired with a real-time reactive user interface built using React and Tailwind CSS.

Milestone 2 Objectives & Scope Alignment

Milestone 2 addresses the critical challenge of converting raw marketplace transactional data into automated merchant intelligence. According to the project brief, this phase is structured around four primary operational capabilities:

- 1. Warehouse Inventory Tracking:** Implementation of warehouse inventory tracking APIs that continuously monitor stock availability and trigger automated low-stock warnings when inventory breaches defined safety thresholds.
- 2. SQL-Based Customer Segmentation:** A dynamic SQL-based customer segmentation engine that evaluates lifetime purchase frequency, total monetary expenditure, and transaction recency to group buyers into actionable behavioral cohorts.
- 3. Algorithmic Recommendations:** An algorithmic rule-based recommendation framework that generates category bestsellers, co-purchasing cross-sell pairings, and customer-affinity-weighted product feeds.
- 4. Demand Run-Rate Forecasting:** A predictive time-series demand forecasting model that calculates daily sales velocity to project 30-day inventory exhaustion timelines and recommend exact restock quantities.

Decoupled Service Architecture & Vendor Data Isolation

The ShopSense architecture enforces a strict three-tier design pattern: the Controller/Router Layer, the Domain Service Layer, and the Data Persistence Layer.

API Routers (app/routers/) serve strictly as lightweight HTTP entry points responsible for route matching, Pydantic schema validation, dependency injection, and HTTP status codes.

All core business logic is encapsulated within dedicated Service Layer modules (app/services/):

- **InventoryService:** Manages stock aggregation, safety threshold filtering, transactional restock operations, and demand velocity run-rate calculations.
- **CustomerAnalyticsService:** Executes SQL aggregation queries computing customer lifetime value, average order values, and RFM segmentation tiers.
- **RecommendationService:** Applies heuristic scoring algorithms to determine category sales velocity rankings, cross-sell affinity pairs, and customer affinity weightings.

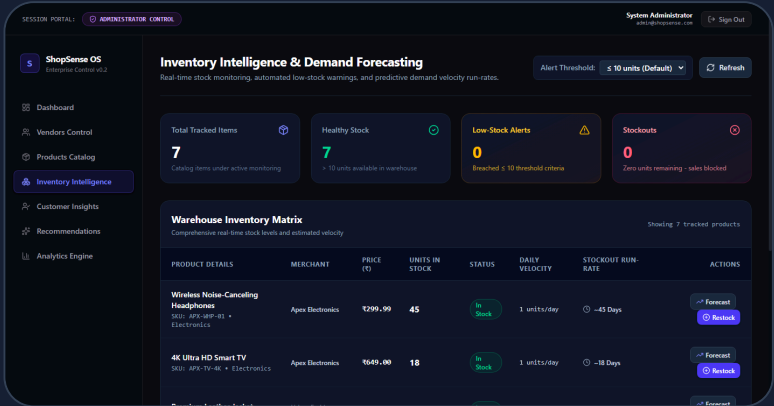
Role-Based Access Control ensures complete data isolation: platform administrators access marketplace-wide aggregate analytics, while vendor merchants access analytics strictly scoped to their store (vendor_id).

Real-Time Warehouse Stock Tracking & Safety Stock Governance

The inventory tracking engine continuously audits warehouse catalog items across three distinct operational states: In Stock, Low Stock, and Out of Stock. Stock levels are tracked in real time against configurable safety thresholds (5, 10, 15, or 20 units).

When a product's available units breach the threshold, the system immediately surfaces a Critical Action Required banner on the dashboard, isolating at-risk inventory before stockouts can disrupt storefront sales.

Each tracked catalog item is mapped to its active sales velocity, calculated by dividing cumulative fulfilled units by product lifecycle days. This velocity produces a real-time Stockout Run-Rate estimate, alerting merchants exactly how many days of supply remain before warehouse depletion.

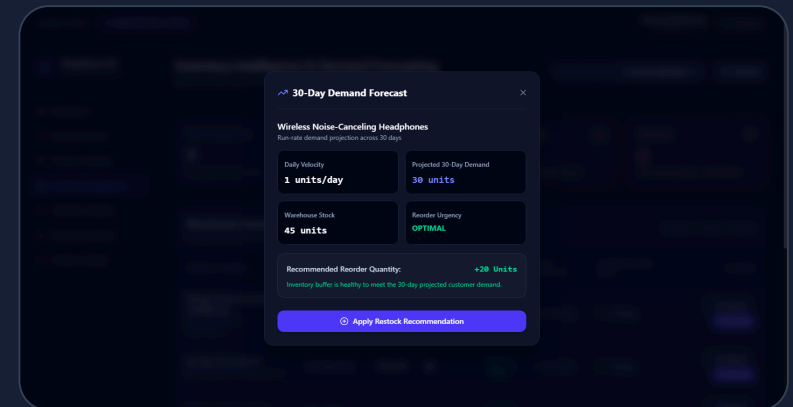


30-Day Predictive Demand Forecasting & Atomic Restock Engine

To eliminate reactive restocking and supply chain bottlenecks, ShopSense incorporates a 30-day predictive demand run-rate forecasting model. The forecasting algorithm analyzes daily sales velocity, calculates projected customer demand over a 30-day horizon, and compares it against current physical inventory.

The system generates a Reorder Urgency classification—flagging items as Immediate, Upcoming, or Optimal—and outputs an exact Recommended Reorder Quantity designed to maintain safety buffers without overstocking capital.

Merchants can execute immediate inventory replenishment directly from the forecasting interface using a dedicated Restock modal (PUT /api/v1/inventory/restock). Restock transactions execute atomically, instantly updating stock quantities, transitioning product status pills from Low Stock to In Stock, and recalculating exhaustion timelines across the platform.

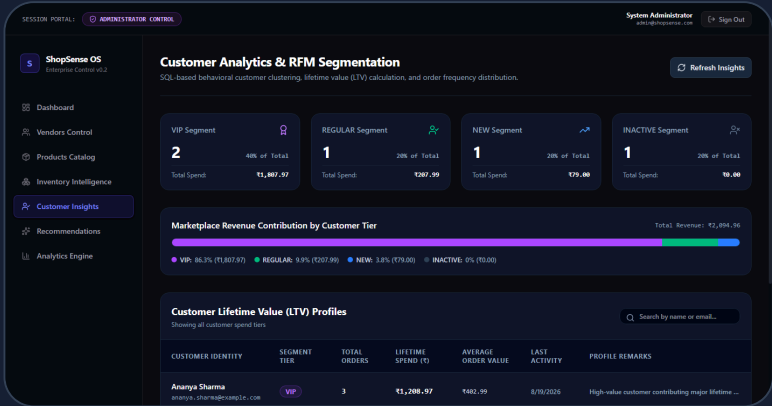


SQL-Based Customer Segmentation & RFM Behavioral Clustering

Customer segmentation in ShopSense is dynamically executed via SQL aggregation in CustomerAnalyticsService, evaluating customer purchasing history across Recency, Frequency, and Monetary (RFM) dimensions. Customers are automatically classified into four distinct cohorts:

- **VIP Cohort:** High-value buyers contributing major lifetime revenue (lifetime expenditure ≥ 500 or ≥ 5 completed orders).
- **Regular Cohort:** Consistent repeat customers demonstrating steady transaction frequency (expenditure between 150 and 500).
- **New Cohort:** Introductory or single-order buyers with lifetime spend under 150.
- **Inactive Cohort:** Registered customer profiles with zero completed transactions.

The platform calculates aggregate metrics per cohort, including customer headcount percentages, total revenue generated, and segment-specific Average Order Value (AOV), providing clear visibility into buyer retention and revenue distribution.

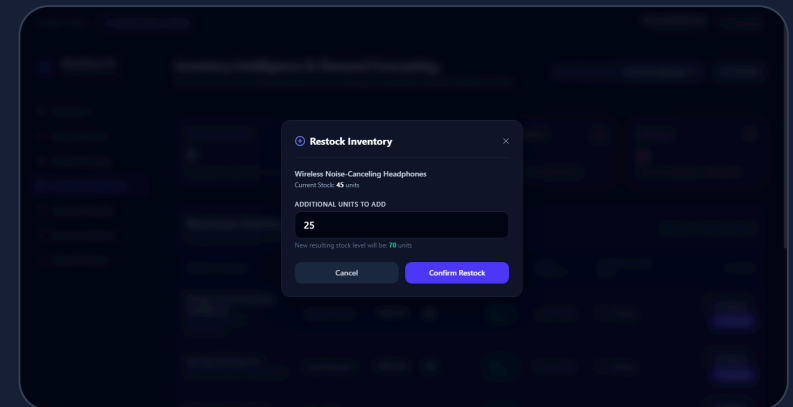


Customer Lifetime Value (LTV) Analysis & Vendor-Scoped Insights

To support strategic customer relationship management, the Customer Insights module features a visual Revenue Contribution breakdown that illustrates the proportion of total sales volume driven by each customer tier.

Complementing the macro-level distribution is an interactive Customer Lifetime Value (LTV) Profiles table. This interface enables users to inspect individual buyer metrics, including total completed orders, lifetime expenditure, average order value, date of last purchase activity, and automated behavioral remarks.

The customer analytics engine dynamically adapts based on user role: platform administrators inspect marketplace-wide aggregate buyer profiles, while approved vendor merchants access customer insights strictly scoped to buyers who have purchased from their specific store (vendor_id), preserving vendor data isolation while providing store-level customer intelligence.

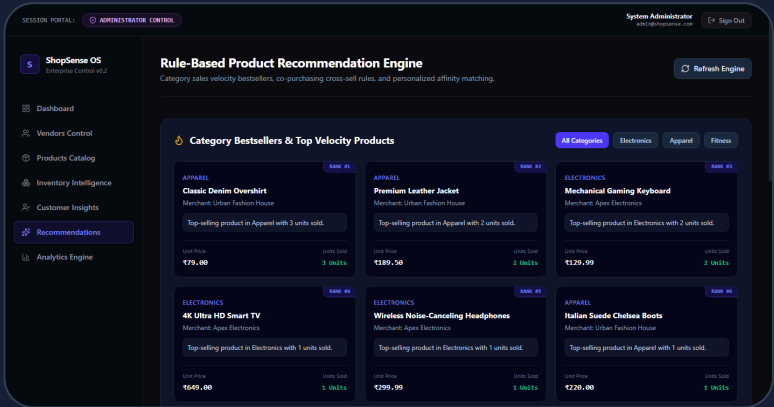


Category Bestseller Recommendation Framework

The ShopSense recommendation engine implements heuristic algorithmic models to identify top-performing products across the marketplace. The Category Bestseller module continuously evaluates all catalog items within specific product categories (such as Electronics, Apparel, and Fitness) or across the platform as a whole.

Products are ranked using an algorithmic scoring function that weights historical completed units sold, active stock availability factors, and unit price normalization.

Each recommendation card displays the product's marketplace rank, merchant storefront name, current unit price, cumulative units sold, and an explicit algorithmic rationale explaining why the item was selected (for example, "Top-selling product in Apparel with 3 units sold").

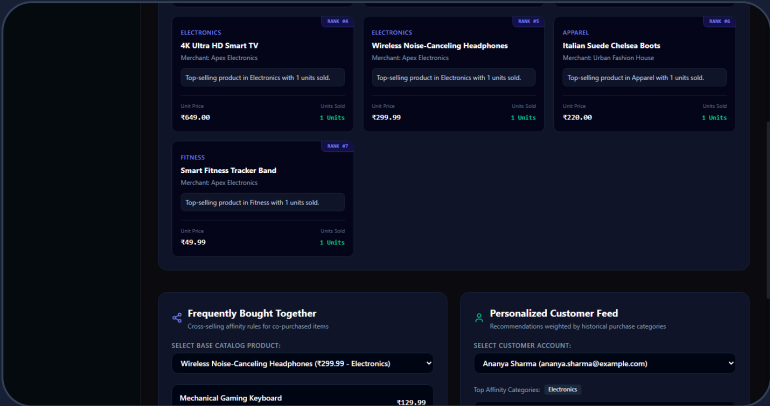


Cross-Sell Rules & Personalized Customer Feeds

The recommendation suite is further expanded through two specialized algorithmic modules:

Frequently Bought Together Cross-Sell Engine: When a user inspects a base catalog item, the algorithm evaluates complementary items from the same category or merchant store, computing a co-purchase affinity score and filtering out depleted inventory.

Personalized Customer Recommendation Feed: This module inspects an individual customer's historical order records to determine their top affinity categories. The engine scores untracked marketplace products by applying category preference multipliers, producing a tailored product feed aligned with the buyer's past purchasing patterns.



OpenAPI Documentation & Milestone Summary

Milestone 2 successfully fulfills all functional, analytical, and architectural objectives required for this phase:

- 1. REST API Documentation:** All endpoints under `/api/v1/inventory/`, `/api/v1/customer-analytics/`, and `/api/v1/recommendations/` are fully documented via FastAPI's interactive OpenAPI 3.0 Swagger UI specification (<http://127.0.0.1:8000/docs>).
- 2. Role-Based Data Isolation:** Dual-token HTTP-Only cookie authentication and RBAC ensure vendor catalog isolation and scoped customer analytics.
- 3. Decoupled Architecture:** Clean separation between FastAPI routers and core business logic in Python service modules (`app/services/`).
- 4. Production Build:** The React frontend compiles cleanly with zero linting or bundle errors, establishing a solid foundation for Milestone 3 Business Intelligence and Advanced Reporting.

